A few other open source tools such as CircleCI, TravisCI and TeamCity are used solely for continuous integration activities.

Continuous monitoring open source tools differ from each other in terms of scalability, availability, amount of data, real-time metric handling capacity and user interface. Examples of a few Windows monitoring open source tools are Prometheus, Grafana, New Relic and Nagios. Windows open source tools for continuous security, such as Snyk and Netsparker, help find and fix vulnerabilities automatically. Open source logging stacks like ElasticSearch, LogStash and Kibana (ELK stack) are used to create a clear aggregation and understanding of log files. Many of these tools mentioned are interoperable, and many enterprises choose from a wide repertoire of these solutions for DevOps.

Containerisation and OSS tools

Containerisation has become very prevalent today in the age of cloud. It is an alternative or a companion to virtualisation, allowing encapsulation and packing-up of software code and all its dependencies, configuration files and libraries so that it can run on any infrastructure. It can be seen as an OS-level virtualisation used to deploy and run distributed applications without launching a separate virtual machine for each application. Its popularity is increasing due to its light weight property, resulting in less resource consumption, greater cost savings, efficiency and higher security. In the DevOps world, containerisation accentuates the aspects of integrating portability with continuous delivery of software releases, making the lives of developers and infrastructure engineers easier. The most popular open source tool that supports containerisation is Docker. Many open source containerisation orchestrators like Kubernetes, Docker Swarm, and Apache Mesos support management and orchestration of software containers.

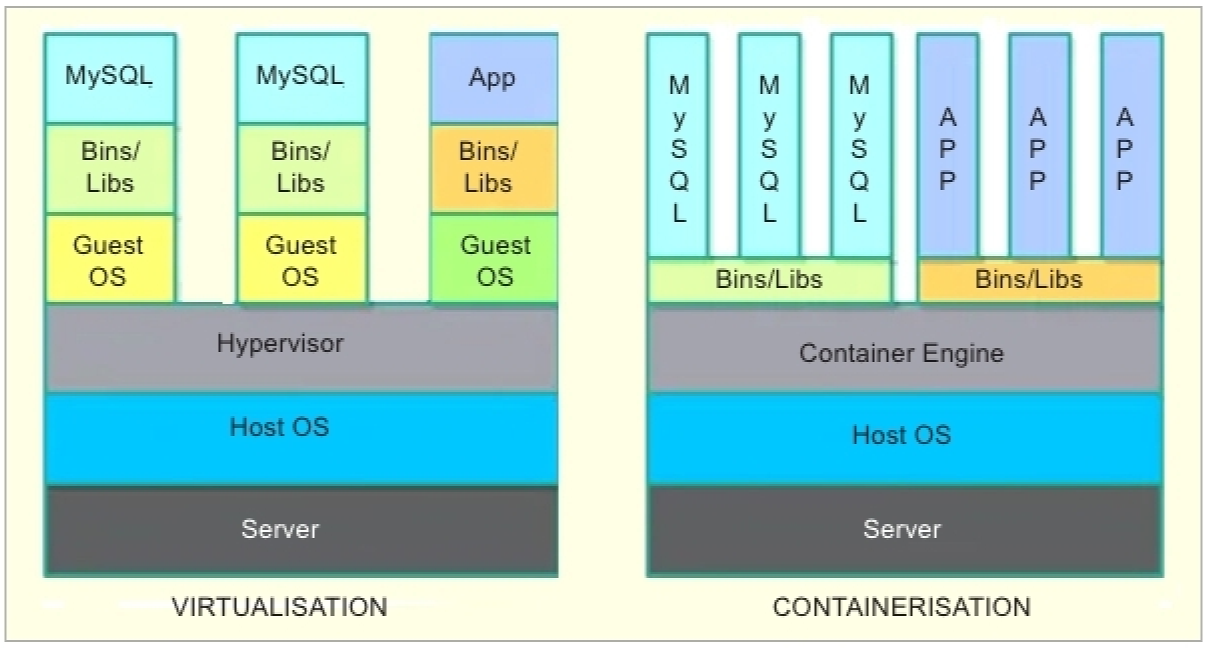

Figure 3: Virtualisation and containerisation

DevOps project management and OSS tools

From the project management point of view, the DevOps product development lifecycle is managed very much like other agile methodologies. The emphasis in DevOps is on automation of software delivery. DevOps projects are handled like any Agile Scrum project, with daily stand-up calls facilitated by a Scrum master. The product owner maintains all the work items and buckets the work into sprints that deliver the product incrementally. Some DevOps projects adopt other agile methodologies like Kanban that use boards to visualise the movement of work items.

A variety of open source tools are available for efficient DevOps project management. For example, Jira and Clarizen allow issue or bug creation and tracking, document sharing, task management and project tracking, Slack allows communication and collaboration and Zoom allows Web conferencing. Most of these tools have customisable and user-friendly interfaces, and integrate with one another.

Finding the right combination of tools is the first step towards a well-designed software solution. DevOps teams prefer to use open source tools to avoid vendor lock-in that is caused by using proprietary stacks exclusively. It helps avoid risks such as vendors ending their support or changing the pricing models. The popularity of these tools allows new members to be on-boarded easily and ensures that the DevOps teams are actively learning and upskilling themselves. END

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